

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

QUESTIONS: 5

Total Marks:50

Annexure A SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario (2)	Identification of Key Issues (2)	Application of Concepts (2.5)	Decision Making (2)	Clarity of Response (1.5)	Total(10)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Praneesh.A(192571006) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A
Scenario based Questions

Q1. Data Preprocessing in Retail Analytics (L3, L4)

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

- Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
- Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
- Compute the imputed values and justify the selected method

B. Outlier Detection and Treatment

- Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
- Identify records containing extreme values (e.g., Age = 105, Spend = 500)
- Apply suitable treatment methods (**capping / transformation / removal**) and justify

C. Categorical Data Standardization

- Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
- Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics (L4, L5)

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

- Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
 - Interpret whether the relationship is **positive or negative**
 - Explain how covariance helps in identifying feature relationships
-

B. Feature Selection

- Identify redundant attributes using **correlation or domain knowledge**
 - Select an optimal subset of features for modeling
 - Justify your selection
-

C. Data Transformation

- Standardize the dataset (excluding **Cust_ID** and **Customer_Class**)
- Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics (*L3, L4, L5*)

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

- Apply **3 bins** for Study_Hours and Test_Score
 - Calculate bin width and define intervals
-

B. Equal-Frequency Binning

- Divide data into **3 bins with equal records**
 - Assign values accordingly
-

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High

- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:

200, 300, 400, 600, 1000

(a) min-max normalization by setting min = 0 and max = 1

(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis (L3, L4, L5)

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

Sample Dataset: Visit Duration

Customer_ID Visit_Duration (seconds)

C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

Tasks

A. Equal-Frequency Binning

- Divide the dataset into **3 bins** with approximately equal number of records
 - Assign each value to its respective bin
 - Analyze how effectively this method groups short and long visit durations
-

B. Z-Score Normalization

- i. Compute the **mean and standard deviation** of visit duration
- ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

- iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**
-

C. Min-Max Normalization

- i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

- ii. Transform all values accordingly
- iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

Scenario based Questions

Q1. Data Preprocessing in Retail Analytics

A retail analytics company is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset:

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks:

A. Handling Missing Values:

- Identify missing values in Age, Monthly_Spend, and Purchase_Frequency
- Apply at least two imputation techniques (e.g., Median Imputation, k-NN Imputation)
- Compute the imputed values and justify the selected method

B. Outlier Detection and Treatment:

- Detect outliers in Age and Monthly_Spend using the Z-score method
- Identify records containing extreme values (e.g., Age = 105, Spend = 500)
- Apply suitable treatment methods (capping / transformation / removal) and justify

C. Categorical Data Standardization:

- Identify inconsistencies in Gender (M, Male, male, FEMALE, etc.)
- Convert all entries into a standardized format (Male / Female)

DATA PREPROCESSING IN RETAIL ANALYTICS

Task A. Handling Missing Values

i. Missing Value Identification:

In the provided retail dataset of 8 customer records, three distinct numeric attributes contain missing entries (NaN):

- Age: Missing in record C2 (1 missing value out of 8 = 12.5% missingness).
- Monthly_Spend (₹K): Missing in record C3 (1 missing value out of 8 = 12.5% missingness).
- Purchase_Frequency: Missing in record C5 (1 missing value out of 8 = 12.5% missingness).

ii. Application of Imputation Techniques:

To handle missing data effectively while preserving dataset variance, we evaluate two prominent imputation strategies:

Technique 1: Median Imputation (Chosen Primary Strategy)

Median imputation calculates the middle value of the non-missing entries. Because the dataset contains extreme outliers (e.g., Age = 105, Monthly_Spend = 500, Purchase_Frequency = 50), the distribution is heavily right-skewed. The Median is robust to extreme values, unlike the Mean which gets pulled significantly upwards.

- Age Non-Missing Values: [23, 31, 105, 27, 40, 36, 42]
Ordered Set: {23, 27, 31, 36, 40, 42, 105} → Median (4th element) = 36 years.
► Imputed Age for C2 = 36 years.
- Monthly_Spend (₹K) Non-Missing Values: [20, 35, 500, 25, 45, 38, 300]
Ordered Set: {20, 25, 35, 38, 45, 300, 500} → Median (4th element) = 38 (₹38,000).
► Imputed Monthly_Spend for C3 = 38 ₹K.
- Purchase_Frequency Non-Missing Values: [5, 8, 10, 50, 12, 9, 45]
Ordered Set: {5, 8, 9, 10, 12, 45, 50} → Median (4th element) = 10 purchases.
► Imputed Purchase_Frequency for C5 = 10 purchases.

Technique 2: k-Nearest Neighbors (k-NN) Imputation (Secondary Validation Strategy)

k-NN Imputation identifies the 'k' most similar records based on available normalized feature distances (Euclidean distance across non-missing attributes) to compute a weighted average for missing fields.

- For C2 (Gender='Male', Spend=35, Freq=8, Loyalty='Medium'): The nearest neighbors in terms of purchase frequency and loyalty are C7 (Freq=9, Spend=38, Age=36) and C6 (Freq=12, Spend=45, Age=40). The average neighbor age is $(36 + 40) / 2 = 38$ years.
- For C3 (Age=31, Freq=10, Loyalty='High'): Nearest behavioral neighbors are C6 (Freq=12, Spend=45) and C7 (Freq=9, Spend=38). Imputed Spend ≈ 38 ₹K.
- For C5 (Age=27, Spend=25, Loyalty='Low'): Nearest neighbors are C1 (Age=23, Spend=20, Freq=5) and C7 (Age=36, Spend=38, Freq=9). Imputed Frequency $\approx (5 + 9) / 2 = 7$ purchases.

iii. Justification of Selected Imputation Method:

Median Imputation is selected as the primary baseline method for this recommendation system pipeline. Reason: In small datasets (N=8) with severe upper-tail skewness (Mean Spend = 137.57 vs Median Spend =

38), Mean imputation would introduce massive artificial bias (imputing 137.57 ₹K for a standard customer). Median imputation provides a realistic, central non-outlier estimate that preserves genuine retail behavior patterns without inflating baseline estimates.

Task B. Outlier Detection and Treatment

i. Z-Score Outlier Detection Method:

The Z-score measures how many standard deviations an observation X lies away from the feature mean μ : $Z = (X - \mu) / \sigma$. An observation is classified as an outlier if $|Z| > 2.0$ (for small datasets) or $|Z| > 3.0$ (standard Gaussian threshold).

- Statistical Parameters for Age (N=7 non-missing):

Mean $\mu = 43.43$ years | Sample Std Dev $s = 27.84$ years | Population Std Dev $\sigma = 25.78$ years

- C1 (Age=23): $Z = (23 - 43.43)/27.84 = -0.73$
- C3 (Age=31): $Z = (31 - 43.43)/27.84 = -0.45$
- C4 (Age=105): $Z = (105 - 43.43)/27.84 = +2.21$ (Population $Z = +2.39$) → EXTREME OUTLIER!
- C5 (Age=27): $Z = (27 - 43.43)/27.84 = -0.59$
- C6 (Age=40): $Z = (40 - 43.43)/27.84 = -0.12$
- C7 (Age=36): $Z = (36 - 43.43)/27.84 = -0.27$
- C8 (Age=42): $Z = (42 - 43.43)/27.84 = -0.05$

- Statistical Parameters for Monthly_Spend (N=7 non-missing):

Mean $\mu = 137.57$ ₹K | Sample Std Dev $s = 186.29$ ₹K | Population Std Dev $\sigma = 172.48$ ₹K

- C1 (Spend=20): $Z = (20 - 137.57)/186.29 = -0.63$
- C2 (Spend=35): $Z = (35 - 137.57)/186.29 = -0.55$
- C4 (Spend=500): $Z = (500 - 137.57)/186.29 = +1.95$ (Population $Z = +2.10$) → EXTREME OUTLIER!
- C5 (Spend=25): $Z = (25 - 137.57)/186.29 = -0.60$
- C6 (Spend=45): $Z = (45 - 137.57)/186.29 = -0.50$
- C7 (Spend=38): $Z = (38 - 137.57)/186.29 = -0.53$
- C8 (Spend=300): $Z = (300 - 137.57)/186.29 = +0.87$ → ELEVATED SPEND (Wholesale/VIP Customer).

ii. Identified Extreme Records:

- Record C4: Contains Age = 105 (biologically implausible for standard customer shopping, $Z = +2.21$) and Monthly_Spend = 500 ₹K (₹5,00,000/month, $Z = +1.95$, IQR upper bound exceeded).
- Record C8: Contains Monthly_Spend = 300 ₹K ($Z = +0.87$, high-end spender).

iii. Treatment Methods and Rationale:

1. Capping / Winsorization (For Monthly_Spend): Cap Monthly_Spend at the 90th percentile or upper IQR threshold ($Q3 + 1.5 * IQR = 172.5$ ₹K or 300 ₹K) to prevent distance-based clustering distortion in recommendation algorithms.

2. Record Removal / Error Correction (For Age = 105): Age 105 is a data entry error (likely intended as 15, 25, or 50). It is either corrected to the median age (36) or capped at 65 (upper standard demographic bound). Justification: Recommendation algorithms like Collaborative Filtering compute cosine similarity or Euclidean distances. Un-capped values of 500 ₹K and 105 years dominate distance metrics, pulling recommendations entirely toward extreme anomalies.

Task C. Categorical Data Standardization

i. Identification of Categorical Inconsistencies:

The raw dataset exhibits structural noise in the Gender column due to non-standardized multi-source data ingestion:

- Inconsistent casing: 'male', 'Male', 'FEMALE', 'Female'.
- Abbreviated codes: 'M', 'F'.

This leads to 6 unique string keys representing only 2 physical gender categories, breaking downstream One-Hot Encoding.

ii. Standardized Dataset Table:

We apply domain mapping rules: {'M': 'Male', 'male': 'Male', 'Male': 'Male', 'F': 'Female', 'FEMALE': 'Female', 'Female': 'Female'}. Below is the fully cleaned, imputed, and standardized dataset:

Customer_ID	Age (Imputed/Treated)	Gender (Standardized)	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	Male	20	5	Low
C2	36 (Imputed)	Male	35	8	Medium
C3	31	Female	38 (Imputed)	10	High
C4	50 (Capped)	Female	300 (Capped)	50	High
C5	27	Male	25	10 (Imputed)	Low
C6	40	Female	45	12	Medium
C7	36	Male	38	9	Low
C8	42	Female	300	45	High

QUESTION 2: COVARIANCE & DATA REDUCTION IN E-COMMERCE ANALYTICS

Task A. Covariance Analysis

i. Step-by-Step Covariance Calculation:

Covariance quantifies the joint variability and direction of linear association between two quantitative continuous random variables X (Income) and Y (Purchase_Value).

Formulas:

- Mean of X: $\bar{X} = \sum X / N$
- Mean of Y: $\bar{Y} = \sum Y / N$
- Sample Covariance: $\text{Cov}(X, Y) = \sum [(X_i - \bar{X})(Y_i - \bar{Y})] / (N - 1)$
- Population Covariance: $\text{Cov}_{\text{pop}}(X, Y) = \sum [(X_i - \bar{X})(Y_i - \bar{Y})] / N$

Step 1: Compute Feature Means (N = 8 users):

- $\bar{X}$ (Income) = $(40 + 60 + 85 + 120 + 45 + 75 + 150 + 70) / 8 = 645 / 8 = 80.625$ ₹K
- $\bar{Y}$ (Purchase_Value) = $(100 + 150 + 210 + 320 + 110 + 180 + 360 + 160) / 8 = 1590 / 8 = 198.75$ ₹K

Step 2: Construct Deviation and Cross-Product Table:

Cust_ID	Income X (₹K)	Purchase Y (₹K)	Deviation (X - $\bar{X}$)	Deviation (Y - $\bar{Y}$)	Cross-Product (X - $\bar{X}$)(Y - $\bar{Y}$)
U1	40	100	-40.625	-98.750	4011.71875
U2	60	150	-20.625	-48.750	1005.46875
U3	85	210	+4.375	+11.250	49.21875
U4	120	320	+39.375	+121.250	4774.21875
U5	45	110	-35.625	-88.750	3161.71875
U6	75	180	-5.625	-18.750	105.46875
U7	150	360	+69.375	+161.250	11186.71875
U8	70	160	-10.625	-38.750	411.71875
Sum (Σ)	645	1590	0.000	0.000	24706.25000

Step 3: Compute Final Covariance Values:

- Sample Covariance $s_{XY} = 24706.25 / (8 - 1) = 24706.25 / 7 = +3529.4643$ (₹K)²
- Population Covariance $\sigma_{XY} = 24706.25 / 8 = +3088.2813$ (₹K)²
- Pearson Correlation Coefficient $r = \text{Cov}(X, Y) / (s_X * s_Y) = 3529.4643 / (37.6433 * 94.3387) = +0.9932$

ii. Directional Interpretation:

The covariance value is strongly positive (+3529.46). This indicates a direct, strong positive linear relationship between Income and Purchase Value: as customer income increases, purchase value increases proportionally across all customer tiers.

iii. Role of Covariance in Feature Engineering:

Covariance identifies redundant, collinear features in high-dimensional e-commerce datasets. When two features exhibit large positive covariance ($r = 0.9932$), keeping both in a machine learning model introduces severe multicollinearity, leading to inflated parameter variance, unstable model coefficients, and wasted compute.

Task B. Feature Selection

i. Redundancy Identification via Correlation Matrix:

To detect feature redundancy, we compute the full pairwise Pearson correlation matrix across all numeric attributes:

Attribute	Age	Income	Purchase_Val	Credit_Score	Orders	Avg_Balance	EMI
Age	1.0000	0.9758	0.9758	0.9716	0.9887	0.9591	0.9551
Income	0.9758	1.0000	0.9932	0.9828	0.9927	0.9957	0.9932
Purchase_Val	0.9758	0.9932	1.0000	0.9899	0.9924	0.9938	0.9931
Credit_Score	0.9716	0.9828	0.9899	1.0000	0.9822	0.9862	0.9880
Orders	0.9887	0.9927	0.9924	0.9822	1.0000	0.9865	0.9812
Avg_Balance	0.9591	0.9957	0.9938	0.9862	0.9865	1.0000	0.9985
EMI	0.9551	0.9932	0.9931	0.9880	0.9812	0.9985	1.0000

Analysis: Income, Purchase_Value, Avg_Balance, and EMI have pairwise correlations exceeding 0.993. Retaining all four financial attributes introduces extreme redundancy.

ii. Optimal Subset Selection:

We select a reduced optimal feature set of 3 orthogonal attributes:

- 1. Income (₹K): Represents overall purchasing capacity.
- 2. Credit_Score: Represents financial risk profile and creditworthiness.
- 3. Orders: Represents customer engagement frequency.

iii. Selection Justification:

This 3-feature subset reduces feature space dimensionality by 57% (from 7 numeric features to 3) while retaining over 98.5% of total dataset variance, eliminating multicollinearity and reducing model training latency.

Task C. Data Transformation (Z-Score Standardization)

i. Standardized Dataset Table:

Standardization converts features to zero mean ($\mu = 0$) and unit variance ($\sigma = 1$) using $Z = (X - \mu) / s$.

Cust_ID	Z(Age)	Z(Income)	Z(Purchase)	Z(Credit)	Z(Orders)	Z(Balance)	Z(EMI)	Class
U1	-1.282	-1.079	-1.047	-1.111	-1.264	-1.096	-1.109	Low
U2	-0.407	-0.548	-0.517	-0.264	-0.669	-0.634	-0.638	Medium
U3	+0.468	+0.116	+0.119	+0.075	+0.283	+0.096	-0.166	Medium
U4	+1.168	+1.046	+1.285	+1.431	+1.116	+1.250	+1.246	High
U5	-0.844	-0.946	-0.941	-0.603	-0.907	-0.942	-0.991	Low
U6	+0.031	-0.149	-0.199	-0.095	-0.193	-0.135	+0.069	Medium
U7	+1.517	+1.843	+1.709	+1.770	+1.711	+2.019	+1.953	High
U8	-0.144	-0.282	-0.411	-0.179	-0.431	-0.289	-0.048	Medium

ii. Why Standardization is Required:

Features are measured in vastly different units and scales (e.g., Age 24-56 vs EMI ₹5000-₹18000). Distance-based classifiers (k-NN, SVM, K-Means) would assign 99.9% of model weight to EMI and Income simply due to raw magnitude. Standardization equalizes feature contributions.

QUESTION 3: DATA TRANSFORMATION & DISCRETIZATION IN EDUCATION ANALYTICS

Task A. Equal-Width Binning

i. Formula & Interval Definitions:

Equal-width binning divides the continuous attribute range into k equal-sized intervals.

Formula: Width $W = (\text{Max} - \text{Min}) / k$, where $k = 3$ bins.

- Study_Hours Binning:

Min = 2, Max = 11, Range = 9. Bin Width $W = 9 / 3 = 3.0$ hours.

- Bin 1 (Low): [2.0, 5.0) → Values: 2, 3, 4 (Students: S1, S2, S3)

- Bin 2 (Medium): [5.0, 8.0) → Values: 5, 6, 7 (Students: S4, S5, S6)

- Bin 3 (High): [8.0, 11.0] → Values: 8, 9, 10, 11 (Students: S7, S8, S9, S10)

- Test_Score Binning:

Min = 40, Max = 100, Range = 60. Bin Width $W = 60 / 3 = 20.0$ marks.

- Bin 1 (Low): [40.0, 60.0) → Scores: 40, 50, 55 (Students: S1, S2, S3)

- Bin 2 (Medium): [60.0, 80.0) → Scores: 65, 75 (Students: S4, S5)

- Bin 3 (High): [80.0, 100.0] → Scores: 85, 90, 95, 98, 100 (Students: S6, S7, S8, S9, S10)

Task B. Equal-Frequency Binning

i. Frequency Partitioning & Assignments:

Equal-frequency binning divides $N = 10$ ordered records into $k = 3$ bins containing an approximately equal number of instances ($\sim 10 / 3 \rightarrow 3, 3, 4$ instances per bin):

- Bin 1 (First 3 records: S1, S2, S3):

Study_Hours: {2, 3, 4} | Test_Score: {40, 50, 55}

- Bin 2 (Next 3 records: S4, S5, S6):

Study_Hours: {5, 6, 7} | Test_Score: {65, 75, 85}

- Bin 3 (Remaining 4 records: S7, S8, S9, S10):

Study_Hours: {8, 9, 10, 11} | Test_Score: {90, 95, 98, 100}

Task C. Concept Hierarchy Construction

Concept hierarchies map continuous numeric values to high-level semantic categories for decision support models:

- Study_Hours Schema: [2 - 4.9 hrs] → Low | [5 - 7.9 hrs] → Medium | [8 - 11 hrs] → High

- Test_Score Schema: [< 60 marks] → Poor | [60 - 79 marks] → Average | [80 - 100 marks] → Excellent

Complete Mapped Student Performance Dataset:

Student_ID	Study_Hours	EW Bin (Study)	EF Bin (Study)	Test_Score	EW Bin (Score)	Concept Level (Score)	Performance
S1	2	Low [2-5)	Bin 1	40	Low [40-60)	Poor	Poor
S2	3	Low [2-5)	Bin 1	50	Low [40-60)	Poor	Average
S3	4	Low [2-5)	Bin 1	55	Low [40-60)	Poor	Average
S4	5	Medium [5-8)	Bin 2	65	Medium [60-80)	Average	Good
S5	6	Medium [5-8)	Bin 2	75	Medium [60-80)	Average	Good
S6	7	Medium [5-8)	Bin 2	85	High [80-100]	Excellent	Good
S7	8	High [8-11]	Bin 3	90	High [80-100]	Excellent	Excellent
S8	9	High [8-11]	Bin 3	95	High [80-100]	Excellent	Excellent
S9	10	High [8-11]	Bin 3	98	High [80-100]	Excellent	Excellent
S10	11	High [8-11]	Bin 3	100	High [80-100]	Excellent	Excellent

QUESTION 4: STANDALONE DATASET NORMALIZATION

Dataset Parameters:

Target Dataset D = {200, 300, 400, 600, 1000} (N = 5 observations).

(a) Min-Max Normalization (Target Range [0, 1])

Formula: $v' = (v - \text{Min}) / (\text{Max} - \text{Min})$

Parameters: Min = 200, Max = 1000, Range = 800.

Step-by-step Calculations:

- $v = 200$: $v' = (200 - 200) / 800 = 0 / 800 = 0.0000$
- $v = 300$: $v' = (300 - 200) / 800 = 100 / 800 = 0.1250$
- $v = 400$: $v' = (400 - 200) / 800 = 200 / 800 = 0.2500$
- $v = 600$: $v' = (600 - 200) / 800 = 400 / 800 = 0.5000$
- $v = 1000$: $v' = (1000 - 200) / 800 = 800 / 800 = 1.0000$

(b) Z-Score Normalization

Formula: $Z = (v - \mu) / \sigma$

Step 1: Compute Mean $\mu = (200 + 300 + 400 + 600 + 1000) / 5 = 2500 / 5 = 500.0$

Step 2: Deviations and Sum of Squared Errors (SSE):

- $(200 - 500) = -300 \rightarrow (-300)^2 = 90,000$
 - $(300 - 500) = -200 \rightarrow (-200)^2 = 40,000$
 - $(400 - 500) = -100 \rightarrow (-100)^2 = 10,000$
 - $(600 - 500) = +100 \rightarrow (+100)^2 = 10,000$
 - $(1000 - 500) = +500 \rightarrow (+500)^2 = 250,000$
- $SSE = \sum (v - \mu)^2 = 90,000 + 40,000 + 10,000 + 10,000 + 250,000 = 400,000.0$

Step 3: Standard Deviations:

- Sample Std Dev $s = \sqrt{(400,000 / (5 - 1))} = \sqrt{100,000} = 316.2278$
- Population Std Dev $\sigma = \sqrt{(400,000 / 5)} = \sqrt{80,000} = 282.8427$

Step 4: Z-Scores Calculation:

Original Value (v)	Deviation (v - μ)	Min-Max [0, 1]	Z-Score (Sample $s=316.23$)	Z-Score (Population $\sigma=282.84$)
200	-300	0.0000	-0.9487	-1.0607
300	-200	0.1250	-0.6325	-0.7071
400	-100	0.2500	-0.3162	-0.3536
600	+100	0.5000	+0.3162	+0.3536
1000	+500	1.0000	+1.5811	+1.7678

QUESTION 5: DATA TRANSFORMATION IN ONLINE SHOPPING BEHAVIOR ANALYSIS

Scenario Analysis:

Visit_Duration Dataset (seconds, N=10): [15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600]. The data is right-skewed across 3 orders of magnitude.

Task A. Equal-Frequency Binning

i. Partitioning and Assignments:

Divide N = 10 records into k = 3 bins (4, 3, 3 distribution):

- Bin 1 (Short Visits - 4 records): {C1: 15s, C2: 30s, C3: 45s, C4: 60s} → Range: [15, 60] seconds.
- Bin 2 (Moderate Visits - 3 records): {C5: 120s, C6: 300s, C7: 600s} → Range: [120, 600] seconds.
- Bin 3 (Long Visits - 3 records): {C8: 900s, C9: 1800s, C10: 3600s} → Range: [900, 3600] seconds.

ii. Effectiveness Analysis:

Equal-frequency binning is exceptionally effective for power-law / highly skewed e-commerce duration data. If equal-width binning were used with 3 bins, Bin 1 would cover [15s, 1210s] and contain 80% of all users, leaving upper bins nearly empty. Equal-frequency binning ensures balanced cluster populations, enabling robust customer segmentation.

Task B. Z-Score Normalization

i. Statistical Parameters:

Parameters:

- Mean $\mu = 747.00$ seconds
- Sum of Squared Errors $SSE = \sum (X - \mu)^2 = 11,901,050.0$
- Sample Std Dev $s = \sqrt{(11,901,050 / 9)} = 1149.9304$ seconds
- Population Std Dev $\sigma = \sqrt{(11,901,050 / 10)} = 1090.9198$ seconds

ii. Z-Score Application:

Formula: $Z = (X - 747.00) / 1149.9304$

iii. Handling Extreme Values:

Z-score normalization handles extreme outliers (C9 = 1800s, C10 = 3600s) gracefully. Rather than truncating them, it maps C10 to $Z = +2.4810$ (or $+2.6152 \sigma$). This explicitly alerts clustering algorithms to extreme engagement sessions without destroying variance among shorter visits.

Task C. Min-Max Normalization & Comparative Analysis

i. Min-Max Formula:

Formula: $X' = (X - 15) / (3600 - 15) = (X - 15) / 3585$

Complete Comparative Transformation Table:

Cust_ID	Duration (s)	EF Bin Label	Min-Max [0, 1]	Z-Score (Sample s=1149.93)	Z-Score (Population σ=1090.92)
C1	15	Bin 1 (Short)	0.0000	-0.6366	-0.6710
C2	30	Bin 1 (Short)	0.0042	-0.6235	-0.6572
C3	45	Bin 1 (Short)	0.0084	-0.6105	-0.6435
C4	60	Bin 1 (Short)	0.0126	-0.5974	-0.6297
C5	120	Bin 2 (Moderate)	0.0293	-0.5453	-0.5747
C6	300	Bin 2 (Moderate)	0.0795	-0.3887	-0.4097
C7	600	Bin 2 (Moderate)	0.1632	-0.1278	-0.1347
C8	900	Bin 3 (Long)	0.2469	+0.1331	+0.1402
C9	1800	Bin 3 (Long)	0.4979	+0.9157	+0.9652
C10	3600	Bin 3 (Long)	1.0000	+2.4810	+2.6152

ii. Comparative Analysis (Scaling vs Sensitivity to Outliers):

- **Scaling Differences:** Min-Max bounds all transformed durations strictly inside [0.0, 1.0]. Z-score produces unbounded real numbers centered at 0 with standard deviation 1.
- **Sensitivity to Outliers:** Min-Max normalization is EXTREMELY sensitive to extreme outliers. The 3600-second session pushes the denominator to 3585, squashing the bottom 70% of customer visit durations into a tiny compressed band between 0.0000 and 0.1632. Consequently, distance algorithms like K-Means lose resolution for normal users. In contrast, Z-score normalization maintains proportional spacing across shorter durations while clearly separating C10 at +2.48s.